5a.

Vstupné dáta: 5 skupín bodov, kde každá skupina obsahuje 50 bodov. Každý bod má x y z

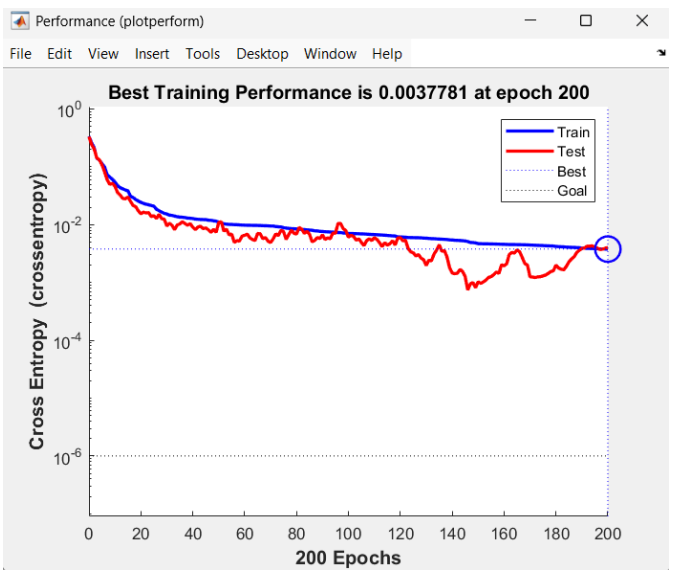
Výstupné dáta: Klasifikácia bodov do 5 skupín.

Rozdelenie dát: 200 tren; 50 validation

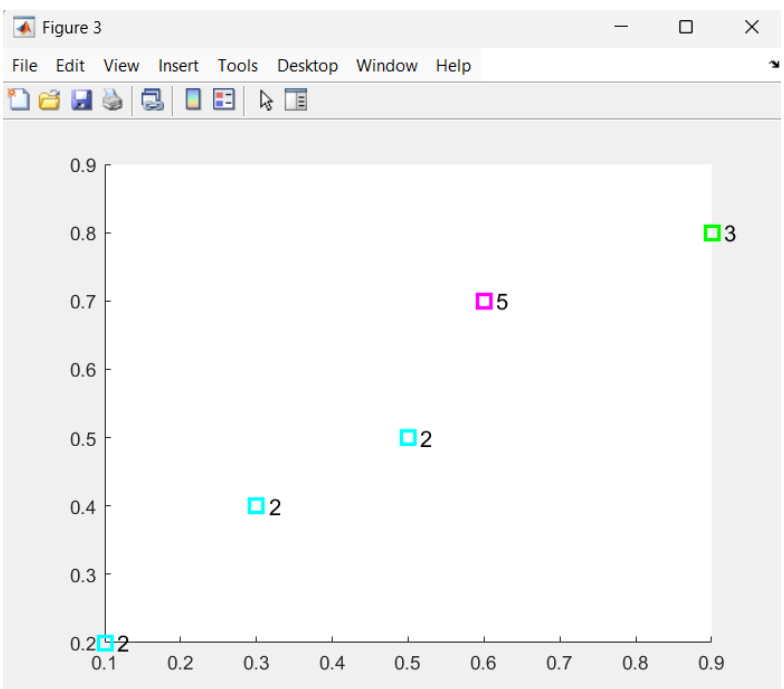
Popis štruktúry MLP siete: vstupy 3 neurony; vystup 5 neuronov; skryta vrstva ma 6 neuronov; patternnet – sigmoid; output layer – softmax (convert to probability)

Cross-Entropy Loss - - kriterialna funkcia

```
% nastavenie parametrov trenovania
net.trainParam.goal = 1e-6; % ukoncovacia podmienka na chybu.
net.trainParam.show = 20; % frekvencia zobrazovania chyby
net.trainParam.epochs = 200; % maximalny pocet trenovacich epoch.
net.trainParam.max_fail=12;
```



Define new points, classify using trained net.

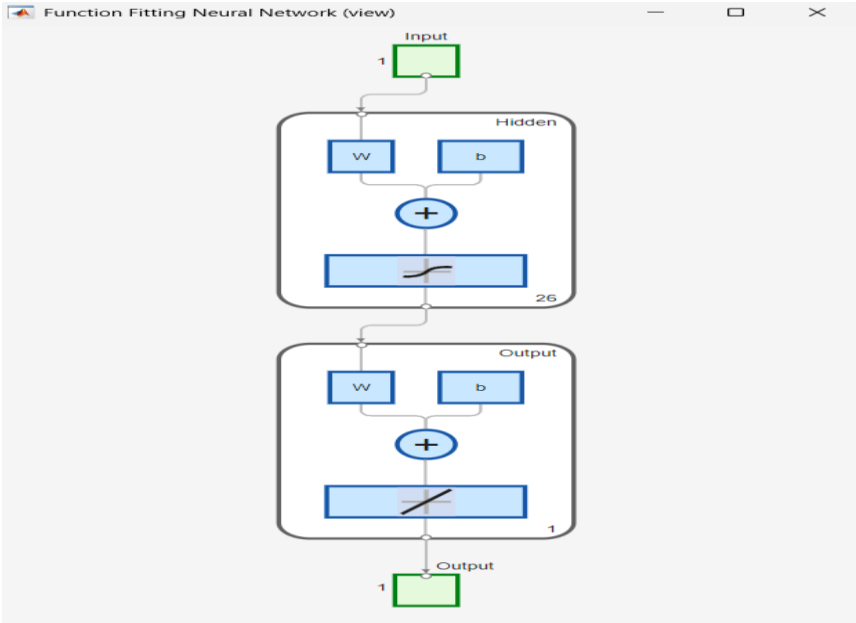


5b.

Uloha: aproximovat nelinearnu funkciu

Vstupné dáta: Jediný vstup zo súboru `datafun.mat`, ktorý obsahuje hodnoty pre x.

Výstupné dáta: Výstupy y, ktoré sú hodnoty cieľovej nelineárnej funkcie pre každý bod x.



Typy:

Input: x

Hiden: tansig (-1; 1)

Ouput: ‘purelin’ (any real number)

Kritériálna funkcia (Loss Function): **Mean Squared Error (MSE)**
Levenberg-Marquardt algorithm

Indexovanie:

net.divideFcn = 'divideind'; % index-based division

net.divideParam.trainInd = indx_train; % train indices

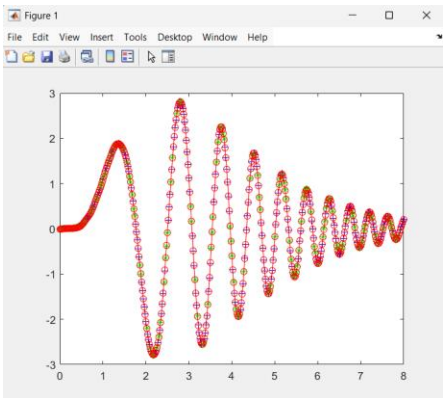
net.divideParam.valInd = []; % no validation data

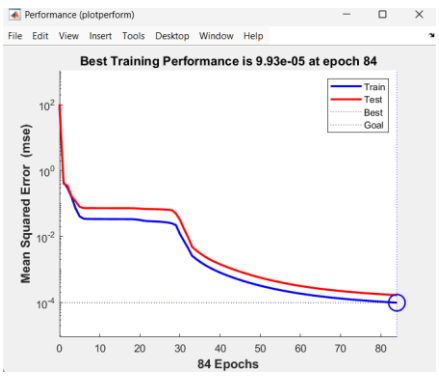
net.divideParam.testInd = indx_test; % test indices

Parametry trenovania:

```
% Nastavenie parametrov trénovania
net.trainParam.goal = 1e-4; % Ukoncovacia podmienka na chybu
net.trainParam.show = 5; % Frekvencia zobrazovania priebehu chyby trénovania
net.trainParam.epochs = 100; % Max. po?et trénovacích cyklov.
net.trainParam.epochs = 100; % maximalny pocet trenovacich epoch.
```

Pocet neuronov co splna kvalitu: 26

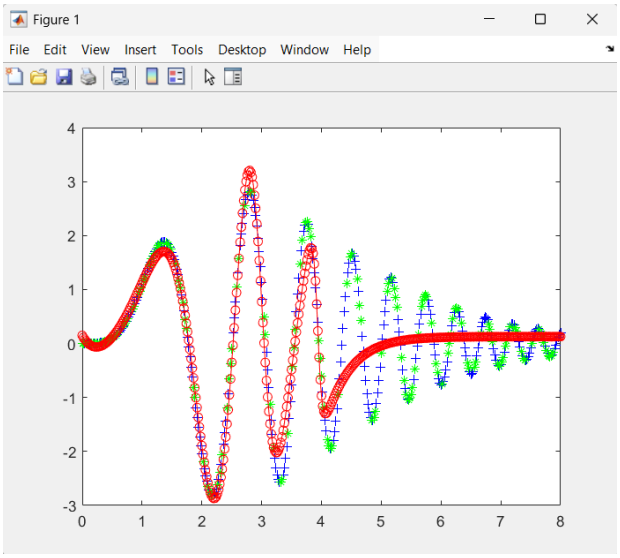




Trénovacie dáta: SSE = 0.023675 MSE = 0.000099 MAE = 0.027633

Testovacie dáta: SSE = 0.028673 MSE = 0.000178 MAE = 0.031156

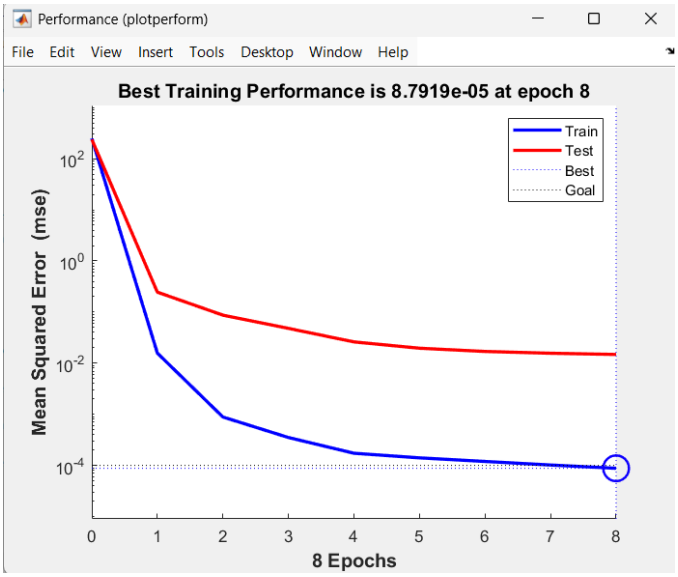
Prilis malo: 5 neuronov

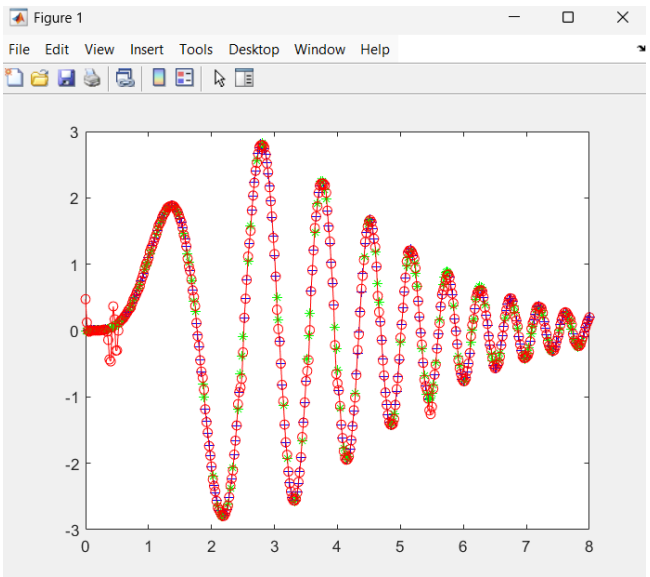


Trénovacie dáta: SSE = 235.610474 MSE = 0.981710 MAE = 2.918329

Testovacie dáta: SSE = 168.381582 MSE = 1.045848 MAE = 2.918407

Prilis vela: 120





Trénovacie dáta: SSE = 0.021101 MSE = 0.000088 MAE = 0.044987

Testovacie dáta: SSE = 2.385382 MSE = 0.014816 MAE = 0.503234

Output:

Blue '+' represents the actual training data (y1).

Green '*' represents the actual test data (y2).

Red circles ('-or') represent the output of the neural network.

6. Uloha:

Úloha: Cieľom bolo klasifikovať kardiozáznamy z CTG do 3 tried: zdravý, podozrivý a patologický

Vstupné dáta: NDATA obsahuje 2126 vzoriek, každá s 25 vlastnosťami

Výstupné dáta: `typ_ochorenia` obsahuje 2126 hodnot, ktoré určujú triedu (1 = zdravý, 2 = podozrivý, 3 = patologický).

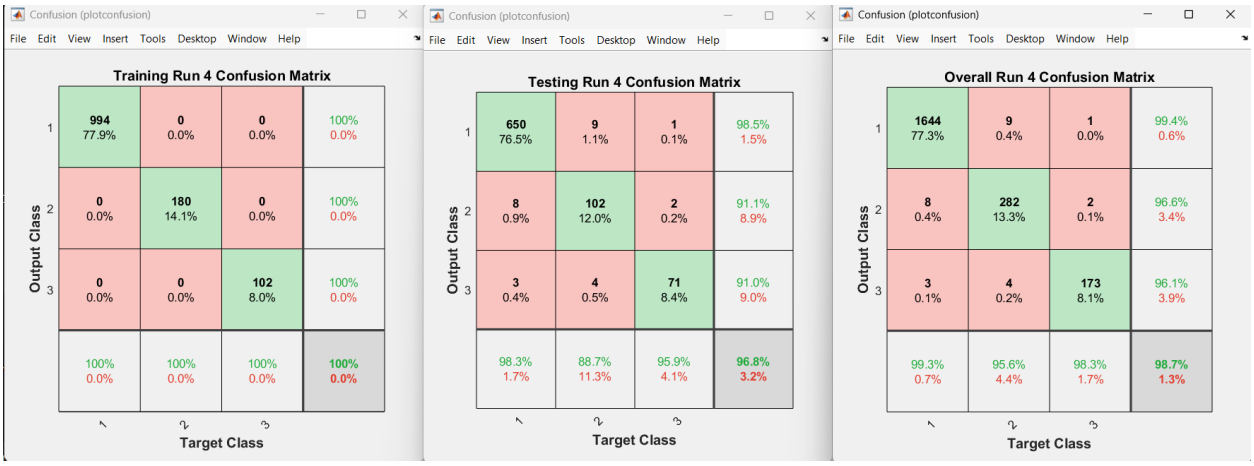
Vstupy: 25 neurónov (pre 25 vlastností CTG).

Skrytá vrstva: 45 neurónov s aktivačnou funkciou **sigmoid**.

Výstupná vrstva: 3 neuróny s **softmax** funkciou (pre 3 triedy).

Ukončovacia podmienka: `net.trainParam.goal = 1e-4`.
Počet epoch: 300.
Použitá kriteriálna funkcia: **Cross-Entropy Loss** (v kombinácii s **softmax** výstupom).
5 runs statistics:
Training Accuracy: The minimum accuracy achieved was 99.84%, the maximum was 100%, and the average was 99.94%.
Test Accuracy: The minimum test accuracy was 92.71%, the maximum was 96.82%, and the average was 94.73%.

Overall Accuracy: The overall accuracy ranged from 97.08% to 98.73%, with an average of 97.86%.



Ine nastavenia:

5 neuronov:
Min Training Accuracy: 93.81 %

Max Training Accuracy: 93.81 %

Avg Training Accuracy: 93.81 %

Min Test Accuracy: 90.94 %

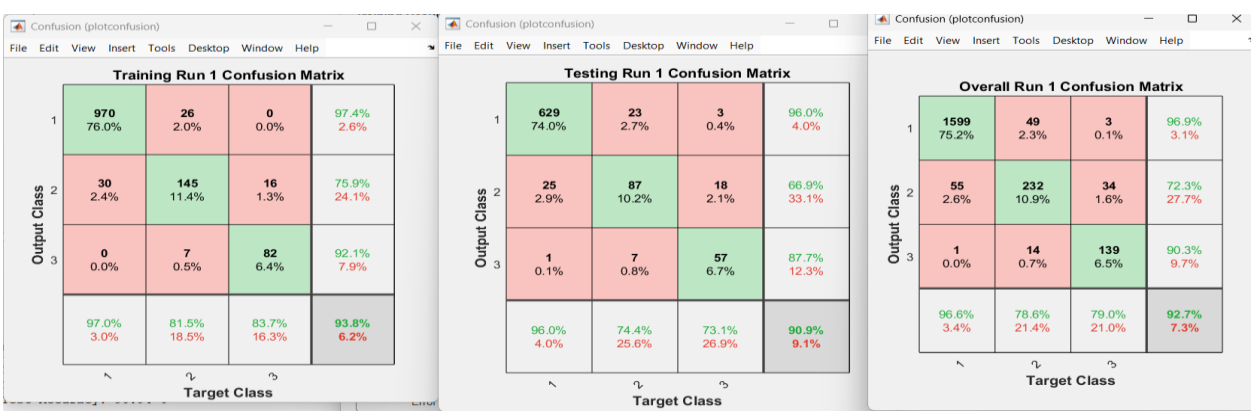
Max Test Accuracy: 90.94 %

Avg Test Accuracy: 90.94 %

Min Overall Accuracy: 92.66 %

Max Overall Accuracy: 92.66 %

Avg Overall Accuracy: 92.66 %



Pre 150 neuronov:
Min Training Accuracy: 99.92 %

Max Training Accuracy: 99.92 %

Avg Training Accuracy: 99.92 %

Min Test Accuracy: 90.94 %

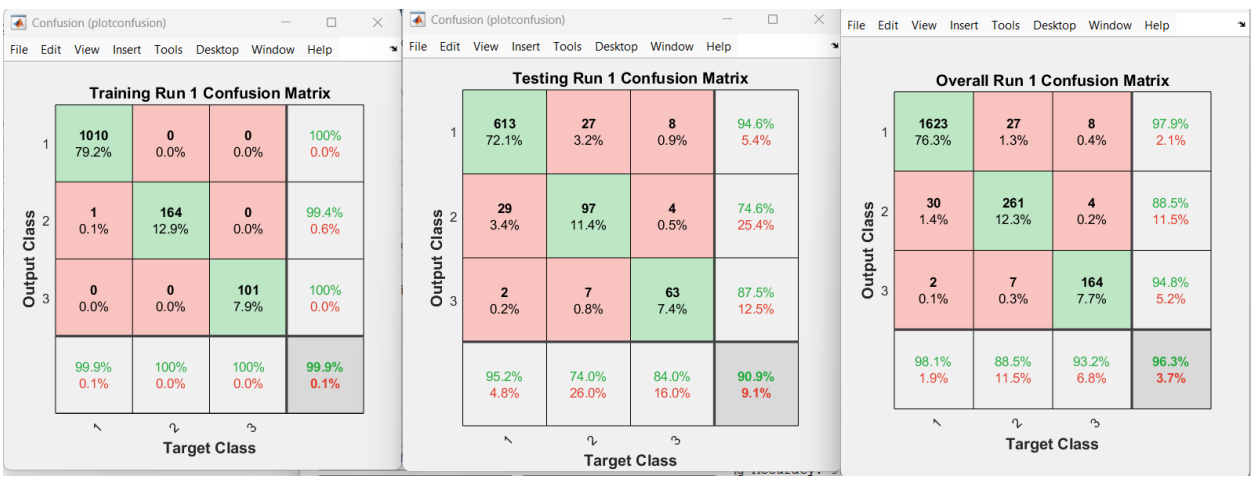
Max Test Accuracy: 90.94 %

Avg Test Accuracy: 90.94 %

Min Overall Accuracy: 96.33 %

Max Overall Accuracy: 96.33 %

Avg Overall Accuracy: 96.33 %



Postup testovanie vybranych vzoriek:

```
% Získaj vstupy a simuluj sieť
selected_inputs = NDATA([sample1, sample2, sample3], :)';
results = sim(net, selected_inputs);
predicted = vec2ind(results);
% Výpis
fprintf('\n--- Klasifikácia jednotlivých vzoriek ---\n');
fprintf('Normálny (1): Predikovaná trieda = %d\n', predicted(1));
fprintf('Podozrivý (2): Predikovaná trieda = %d\n', predicted(2));
fprintf('Patologický (3): Predikovaná trieda = %d\n', predicted(3));
```

1. Group = healthy – negative class

2. and 3. - positive classes

Run 1 - Sensitivity: 96.77 % (model good at detecting positive cases 2,3)

Run 1 - Specificity: 97.58 % (how well model identifies negative cases 1)

7. Uloha
Uloha: natrénujte neurónovú sieť, ktorá bude schopná rozpoznávať rukou písané číslice
Vstupne 784-dimensional vectors ktory reprezentuje 28x28 cislo

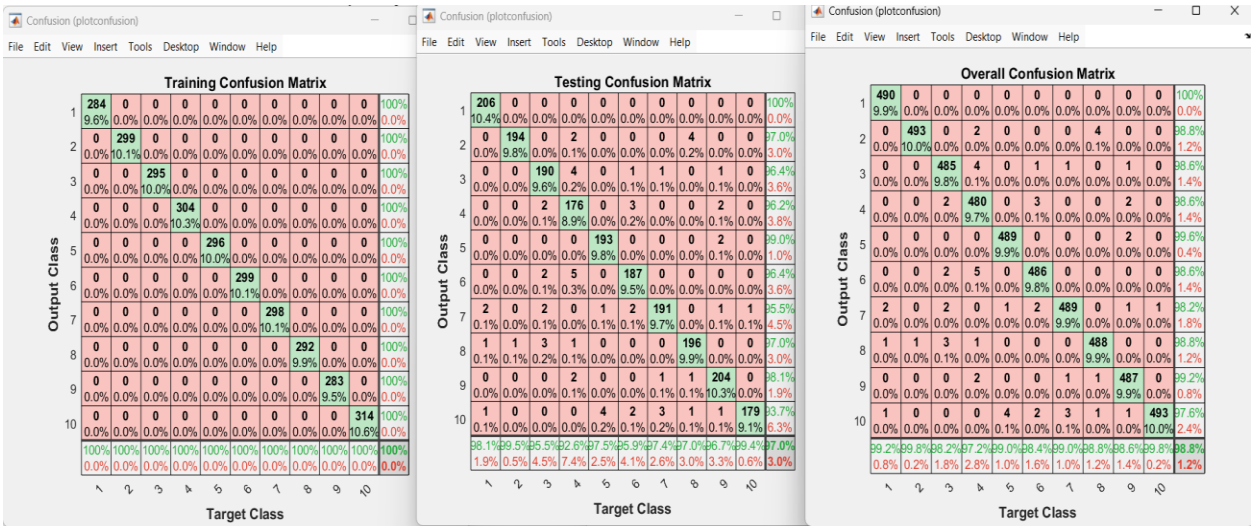
Vystup: zatriedenie do 10 skupin (cisla 0 – 9)
Nahlepsia siet 80 neuronov
60 tren 40 test

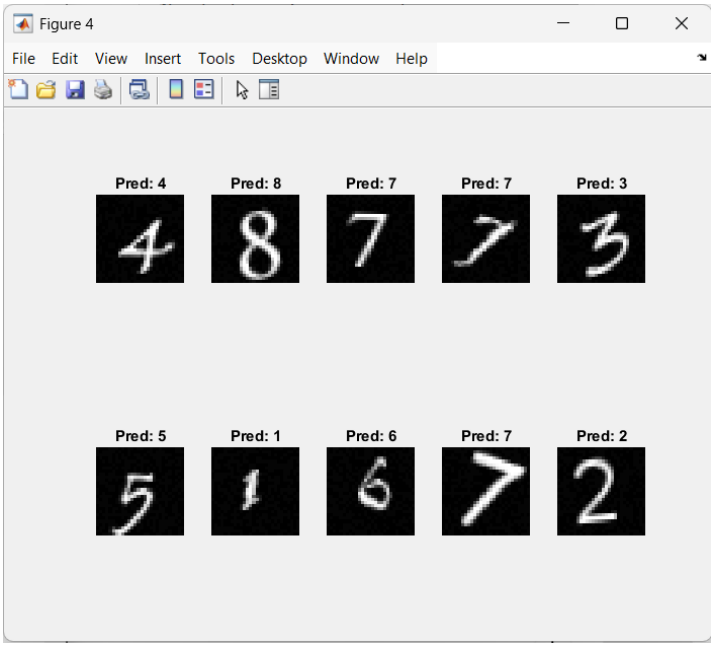
Popis struktury:

Input 784; Hidden: 80: Output: 10. Aktivacna: sigmoid alebo tansig (patternet)

Parametre trenovanie:
Chyba: 1e-5; Epochy: 300: Funkcia Levenberg-Marquardt

Najlepsia:





--- Štatistiky z 5 bežov ---

Minimálna úspešnosť na trénovacích dátach: 100.00 %

Maximálna úspešnosť na trénovacích dátach: 100.00 %

Priemerná úspešnosť na trénovacích dátach: 100.00 %

Minimálna úspešnosť na testovacích dátach: 94.64 %

Maximálna úspešnosť na testovacích dátach: 96.96 %

Priemerná úspešnosť na testovacích dátach: 95.77 %

Minimálna celková úspešnosť: 97.85 %

Maximálna celková úspešnosť: 98.79 %

Priemerná celková úspešnosť: 98.31 %

--- Klasifikácia jednej vzorky z každej číslice (0–9) ---

Číslica 0 → Predikovaná trieda = 0

Číslica 1 → Predikovaná trieda = 1

Číslica 2 → Predikovaná trieda = 2

Číslica 3 → Predikovaná trieda = 3

Číslica 4 → Predikovaná trieda = 4

Číslica 5 → Predikovaná trieda = 5

Číslica 6 → Predikovaná trieda = 6

Číslica 7 → Predikovaná trieda = 7

Číslica 8 → Predikovaná trieda = 8

Číslica 9 → Predikovaná trieda = 9

8. Uloha

V tejto úlohe sme mali vytvoriť a natrénovať konvolučnú neurónovú sieť (CNN) na rozpoznávanie rukou písaných číslic.
60% trénovacích dát a 40% testovacích dát.

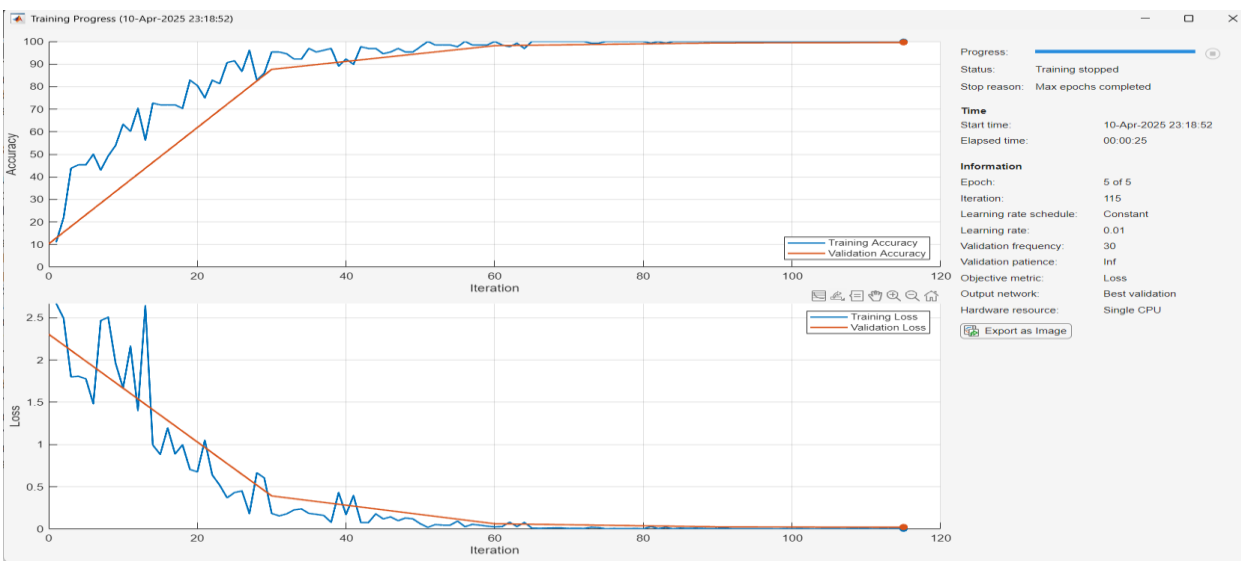
Vstup 784x4940; Vystup 4940x10;
Popis struktury:
Layer type: `imageInputLayer; Input size: [28 28 1]`
Layer type: `convolution2dLayer(5 20); convolution2dLayer(3, 40)`
Layer type: `batchNormalizationLayer (stabilizuje ucenie)`
Layer type: `reluLayer (RuLU funkcia ktora pridava nelinearitu)`
Fully Connected Layer: `transformuje vystup z konvolucneho layer do 1D`
Softmax Layer: `transformuje do pravdepodobnostej`

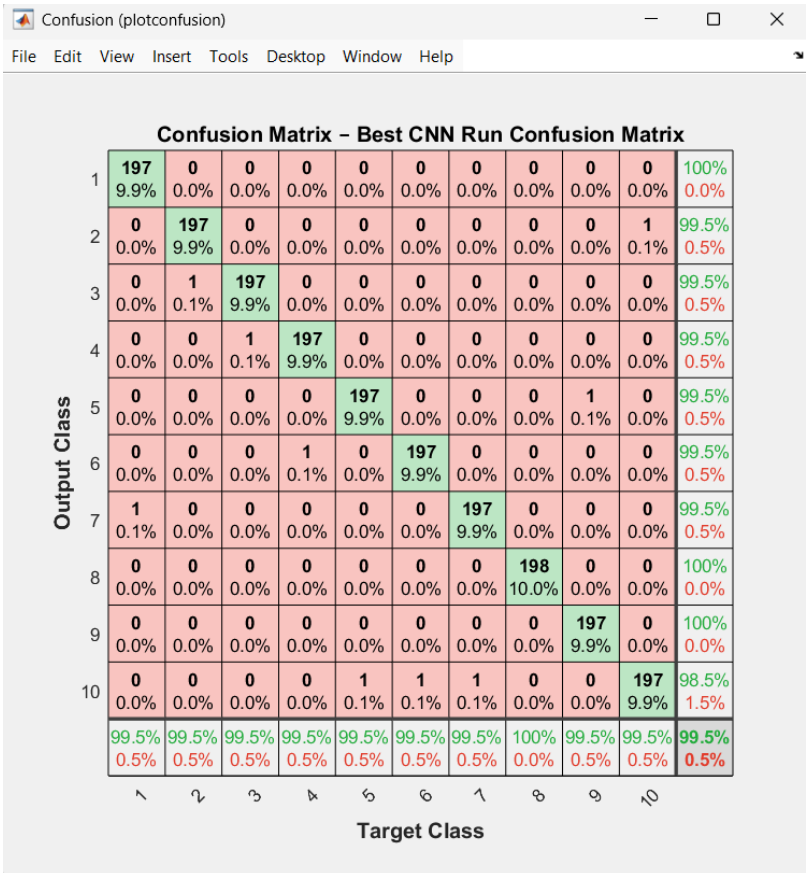
```
Training param:
Optimizer: Stochastic Gradient Descent with Momentum (sgdm).
Epochy: 5
Validacna freq: every 30 iteraci
Loss function: Cross-entropy (hovori ako dobre triedime)
Metrics: Accuracy

Best accuracy from 5 runs: 99.55 %
```

Best Train Accuracy: 100.00 %
Best Validation Accuracy: 99.55 %

Trenovacie:
Min: 100%
Max: 100%:
Average: 100%

Validacia:
Min: 98.99%
Max: 99.55%
Average: 99.24%



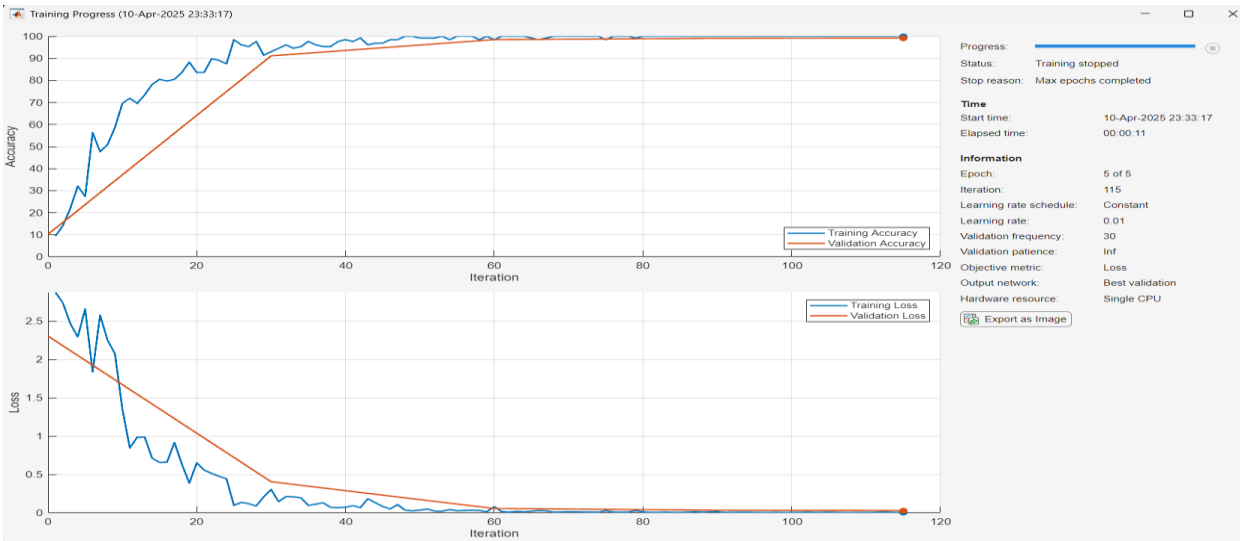


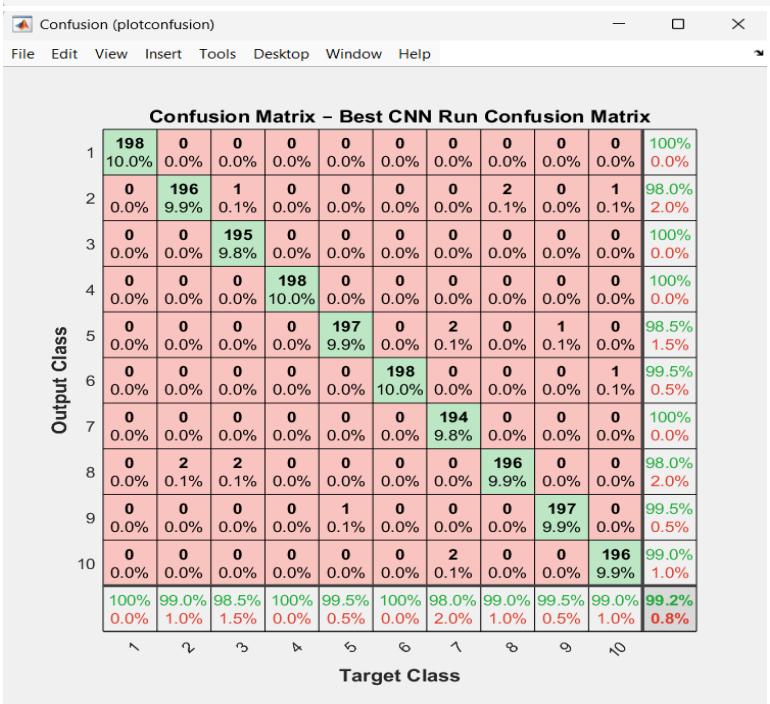
Porovnať dve štruktúry CNN. (removed one convolution layer)

Best accuracy from 5 runs: 99.24 %

Best Train Accuracy: 100.00 %

Best Validation Accuracy: 99.24 %





Output Class \ Target Class	1	2	3	4	5	6	7	8	9	10	
1	**198** 10.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	100% 0.0%
2	**0** 0.0%	**196** 9.9%	**1** 0.1%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**2** 0.1%	**0** 0.0%	**1** 0.1%	98.0% 2.0%
3	**0** 0.0%	**0** 0.0%	**195** 9.8%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	100% 0.0%
4	**0** 0.0%	**0** 0.0%	**0** 0.0%	**198** 10.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	100% 0.0%
5	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**197** 9.9%	**0** 0.0%	**2** 0.1%	**0** 0.0%	**1** 0.1%	**0** 0.0%	98.5% 1.5%
6	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**198** 10.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**1** 0.1%	99.5% 0.5%
7	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**194** 9.8%	**0** 0.0%	**0** 0.0%	**0** 0.0%	100% 0.0%
8	**0** 0.0%	**2** 0.1%	**2** 0.1%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**196** 9.9%	**0** 0.0%	**0** 0.0%	98.0% 2.0%
9	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**1** 0.1%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**197** 9.9%	**0** 0.0%	99.5% 0.5%
10	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**0** 0.0%	**2** 0.1%	**0** 0.0%	**0** 0.0%	**196** 9.9%	99.0% 1.0%
	100% 0.0%	99.0% 1.0%	98.5% 1.5%	100% 0.0%	99.5% 0.5%	100% 0.0%	98.0% 2.0%	99.0% 1.0%	99.5% 0.5%	99.0% 1.0%	**99.2% 0.8%**